CARLOS TIBURCIO

Secure Proxmox Remote Access

Independent Homelab | REMOTE01

Tailscale subnet routing • Linux containers • Access validation

PROJECT GOAL

Manage my Proxmox homelab away from home through a private VPN path, without adding a public port-forward to the management interface.

Result: Proxmox and a Windows VM console worked over a phone hotspot. Access stopped when Tailscale disconnected, returned when it reconnected, and recovered after the gateway container rebooted.

1	/32	8006	4
Dedicated gateway	Single-host route	Permitted TCP port	Validation behaviors

REMOTE01 is a supporting infrastructure project built on my existing Proxmox environment. It improves how I operate the lab and records the design choices, implementation, and tests behind that access.

Completed October 25, 2026. Personal homelab. Container restart recovery was tested; a full Proxmox-host reboot was not.

01 / PROBLEM AND DESIGN

A private path into the lab

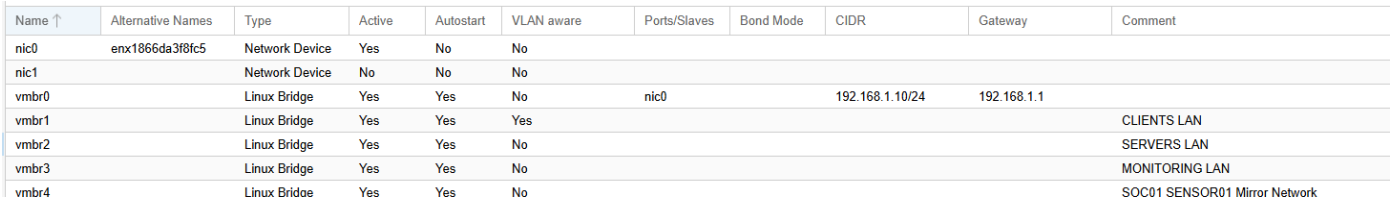
My cybersecurity projects run on a Dell Precision T5810 at home. Local-only management limited when I could work on them. I needed access from another internet connection while keeping Proxmox management off a public router port-forward.

A dedicated OPNsense appliance remains a future home-network upgrade. Tailscale let me solve the immediate access problem using the existing ISP router and server.

Keep the gateway separate from the lab firewall

I created VPN01, an unprivileged Ubuntu LXC container on vmbr0, the bridge carrying Proxmox management. Tailscale runs in this container rather than directly on the hypervisor. The path does not require the lab OPNsense VM to be running.

Component	Configuration	Purpose
Proxmox	192.168.1.10/24 on vmbr0	Management destination
VPN01	Ubuntu 24.04 LXC, CT 500	Tailscale subnet router
carlospc	Windows client with Tailscale	Authorized remote laptop
Home router	192.168.1.1	Existing internet gateway



Name	Alternative Names	Type	Active	Autostart	VLAN aware	Ports/Slaves	Bond Mode	CIDR	Gateway	Comment
nic0	enx1866da3f8fc5	Network Device	Yes	No	No					
nic1		Network Device	No	No	No					
vmbr0		Linux Bridge	Yes	Yes	No	nic0		192.168.1.10/24	192.168.1.1	
vmbr1		Linux Bridge	Yes	Yes	Yes					CLIENTS LAN
vmbr2		Linux Bridge	Yes	Yes	No					SERVERS LAN
vmbr3		Linux Bridge	Yes	Yes	No					MONITORING LAN
vmbr4		Linux Bridge	Yes	Yes	No					SOC01 SENSOR01 Mirror Network

Figure 1. The management bridge uses 192.168.1.10/24 and gateway 192.168.1.1. The lab and malware bridges were not attached to VPN01.

Access path

Remote laptop → Tailscale → VPN01 → Proxmox TCP 8006. Browser VM consoles travel through Proxmox. This allows work inside the lab without granting the laptop direct access to every VM or VLAN.

02 / GATEWAY SETUP

A small container with the device it needs

I downloaded the Ubuntu 24.04 standard LXC template, created CT 500 as VPN01, set a unique root password, and kept the container unprivileged.

Setting	Chosen value
Compute and storage	1 CPU core, 512MB RAM, 512MB swap, 8GB disk on local-lvm
Interface	eth0 on vmbr0, no VLAN tag, firewall checkbox enabled
Addressing	IPv4 DHCP; inherited host DNS; no IPv6 address configured
TUN device	Resources → Add → Device Passthrough → /dev/net/tun

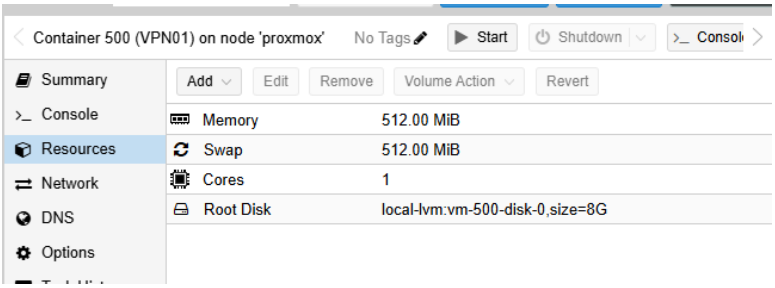


Figure 2. VPN01 resource allocation before adding the TUN device. This screenshot shows the actual 8GB root disk and 512MB memory configuration.

I added /dev/net/tun while the container was stopped, then started it and logged in as root. The initial checks confirmed the device existed, DHCP assigned 192.168.1.161/24, and Proxmox answered three probes with no packet loss.

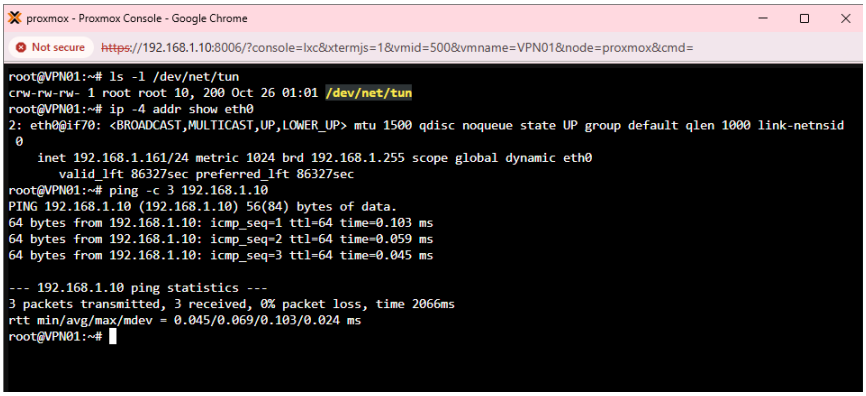


Figure 3. TUN availability and local connectivity were verified before installing Tailscale.

Nesting appears enabled in the recorded container options. Disabling it was recommended but not verified. The documented security boundary is an unprivileged container, not a fully hardened container baseline.

03 / REGISTRATION AND ROUTING

Enroll the gateway and advertise one host

Inside VPN01, I installed Tailscale with its official Linux installer, enabled the daemon at startup, and registered the gateway through the browser sign-in link.

```
apt update
apt install -y curl ca-certificates
curl -fsSL https://tailscale.com/install.sh | sh
systemctl enable --now tailscaled
tailscale up --accept-dns=false
```

The DNS setting preserved VPN01's existing resolver configuration. I installed the Windows client on carlospc and signed in with the same personal account. VPN01 received Tailscale address 100.86.127.70; the laptop received 100.104.158.119.

```
echo 'net.ipv4.ip_forward = 1' > /etc/sysctl.d/99-tailscale.conf
sysctl --system
tailscale set --advertise-routes=192.168.1.10/32
```

I enabled persistent IPv4 forwarding and approved the advertised route in the Tailscale admin console. The /32 route covers only the Proxmox management host. Exit-node mode remained off.

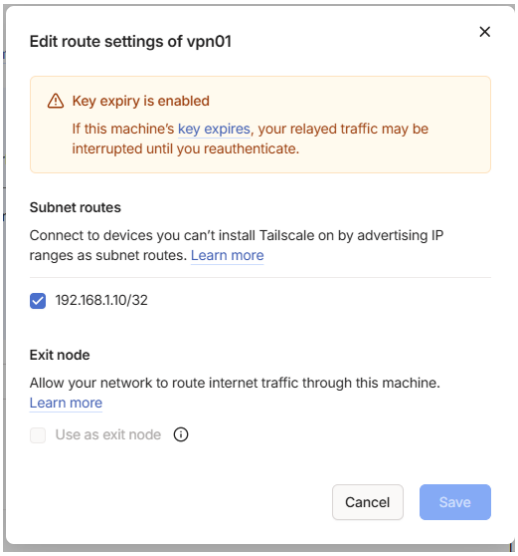


Figure 4. The single-host route is selected. The key-expiry warning was present at this stage; gateway key expiry was disabled afterward.

I disabled key expiry for VPN01 to avoid unattended reauthentication interrupting access, and configured Proxmox Start at boot to Yes for the container. The gateway still depends on protecting the Tailscale account and revoking devices when needed.

04 / ACCESS CONTROL

One source and one management service

The starting policy allowed unrestricted connections. I replaced it with one grant that permits the registered laptop to reach Proxmox on TCP 8006. No broad allow-all rule remained in the replacement policy.

```
{
  "grants": [
    {
      "src": ["100.104.158.119"],
      "dst": ["192.168.1.10"],
      "ip": ["tcp:8006"]
    }
  ]
}
```

Control	Implementation
Source	Laptop Tailscale address 100.104.158.119
Destination	Proxmox management address 192.168.1.10
Route	192.168.1.10/32; no lab VLAN routes advertised
Service	TCP 8006 for the dashboard and browser VM consoles
Authentication	Tailscale device enrollment plus normal Proxmox login
Unpermitted traffic	No allow rule in this Tailscale policy

The source selector is the laptop’s Tailscale address, not its home or hotspot address. A DHCP change on VPN01 does not change the registered gateway’s Tailscale identity. I kept DHCP for the home-facing interface.

What this boundary does and does not cover

These rules apply to traffic through Tailscale. They do not replace local-network firewall rules or Proxmox authentication. Direct laptop connections to Wazuh, RDP, or SSH are outside the configured path.

The working browser console gives broad administrative capability after Proxmox login, so restricting the network path is only one layer. The Proxmox account remains an important part of the access boundary.

05 / EXTERNAL VALIDATION

Work from a different internet connection

I moved the Windows laptop onto my iPhone hotspot and kept Tailscale connected. Proxmox loaded at https://192.168.1.10:8006, and IT-PC01's browser console worked.

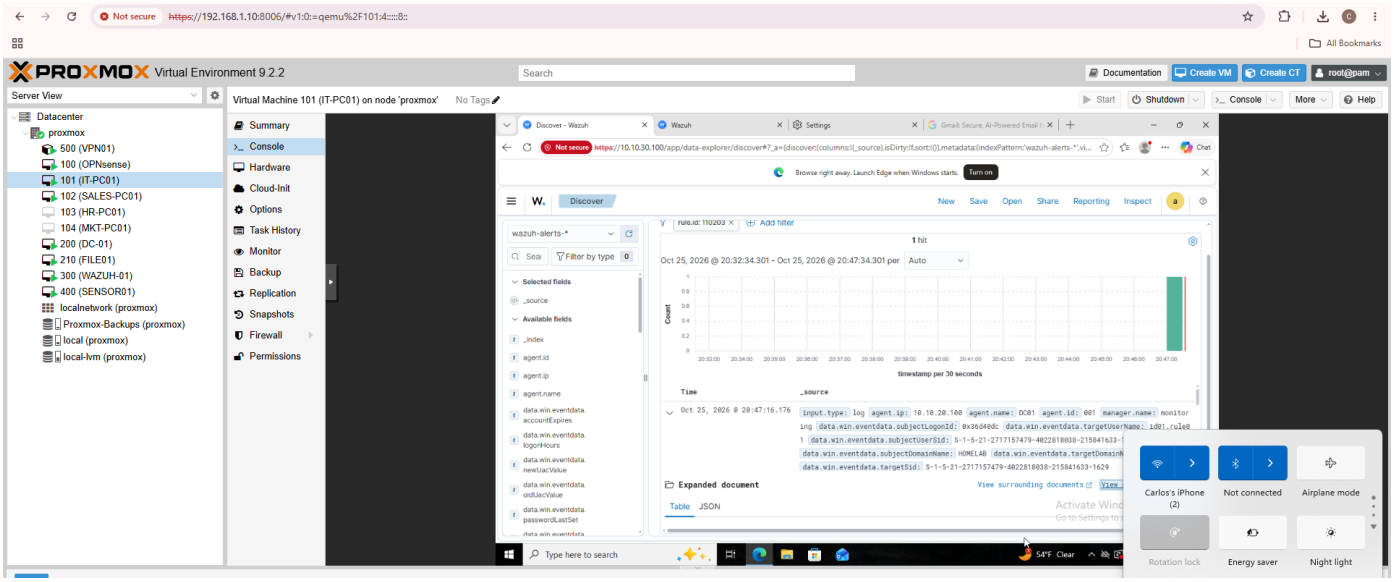


Figure 5. Remote Proxmox access over the iPhone hotspot, with IT-PC01 displaying Wazuh inside the VM.

Why the VM console test matters

A dashboard login alone would not establish that I could continue working on a project. Opening the Windows console confirmed that the workflow I needed was usable through the same permitted management service.

Wazuh at 10.10.30.100 was reached from inside IT-PC01 over the existing lab network. The screenshot does not demonstrate direct access from the remote laptop to the monitoring VLAN.

Condition	Observed result
Laptop on phone hotspot	Separate internet connection used for validation
Tailscale connected	Proxmox dashboard loaded
Browser VM console opened	IT-PC01 was available for lab work

This is the practical outcome of REMOTE01: I can manage the lab and continue working inside its VMs while away from home, without installing a VPN client in each lab guest.

06 / NEGATIVE TEST

Disconnect the VPN and observe the failure

While remaining on the hotspot, I disconnected Tailscale on the laptop. Proxmox lost connectivity. I then reconnected Tailscale and confirmed that management access returned.

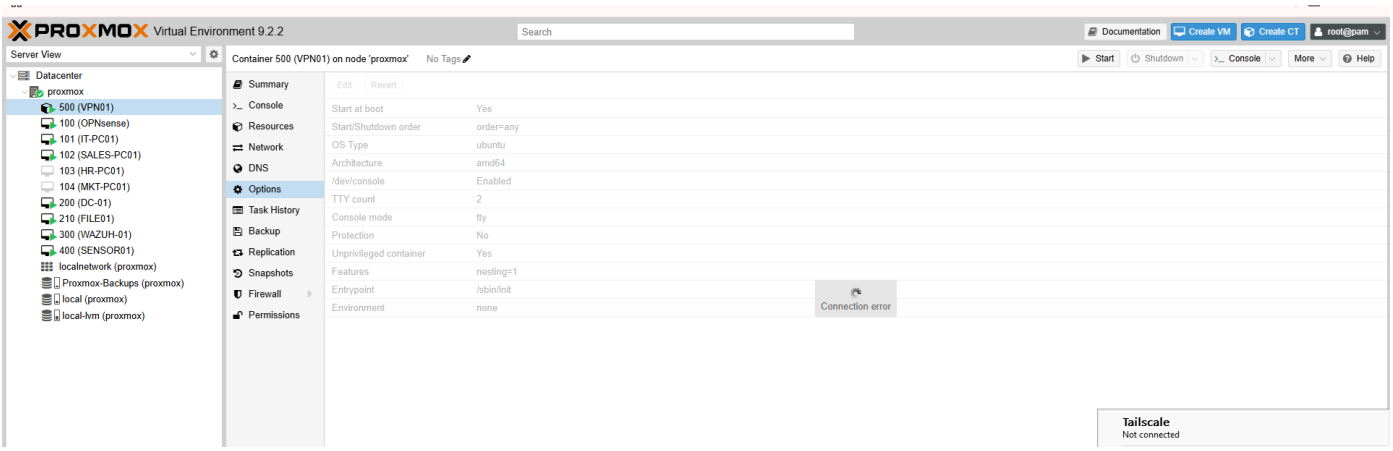


Figure 6. Proxmox reports a connection error while the laptop's Tailscale client shows Not connected. Unrelated account details are outside this crop.

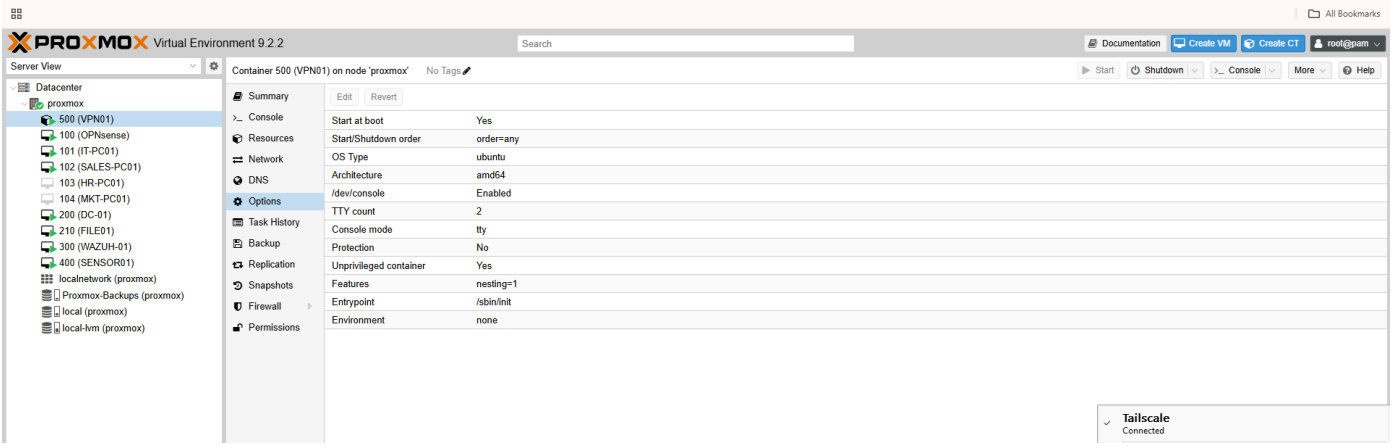


Figure 7. The page returns to its connected state after the laptop reconnects to Tailscale. Start at boot is set to Yes for VPN01.

This test confirmed that the observed remote session depended on the VPN. It was not a separate test of every denied port or an unauthorized device; those remain useful follow-up checks.

Evidence handling

Screenshots are cropped to the relevant interface or result. No settings or test output were changed. Earlier wizard views and duplicate screenshots were omitted so the evidence follows the final configuration and its behavior.

07 / RECOVERY AND OUTCOME

Recover after the gateway restarts

I returned to home Wi-Fi and disconnected Tailscale on the laptop so a local management path remained available. I rebooted VPN01 only, then checked the service, device registration, and forwarding state.

```
systemctl is-active tailscaled
tailscale status
sysctl net.ipv4.ip_forward
```

```
root@VPN01:~# systemctl is-active tailscaled
active
root@VPN01:~# tailscale status
100.86.127.70    vpn01      carlomtj15@  linux    -
100.104.158.119  carlospc   carlomtj15@  windows  offline, last seen 3m ago
root@VPN01:~# systemctl net.ipv4.ip_forward
Unknown command verb 'net.ipv4.ip_forward'.
root@VPN01:~# sysctl net.ipv4.ip_forward
net.ipv4.ip_forward = 1
root@VPN01:~#
```

Figure 8. After reboot, tailscaled is active, VPN01 retains its VPN address, and IPv4 forwarding is 1. An initial systemctl typo was corrected to sysctl.

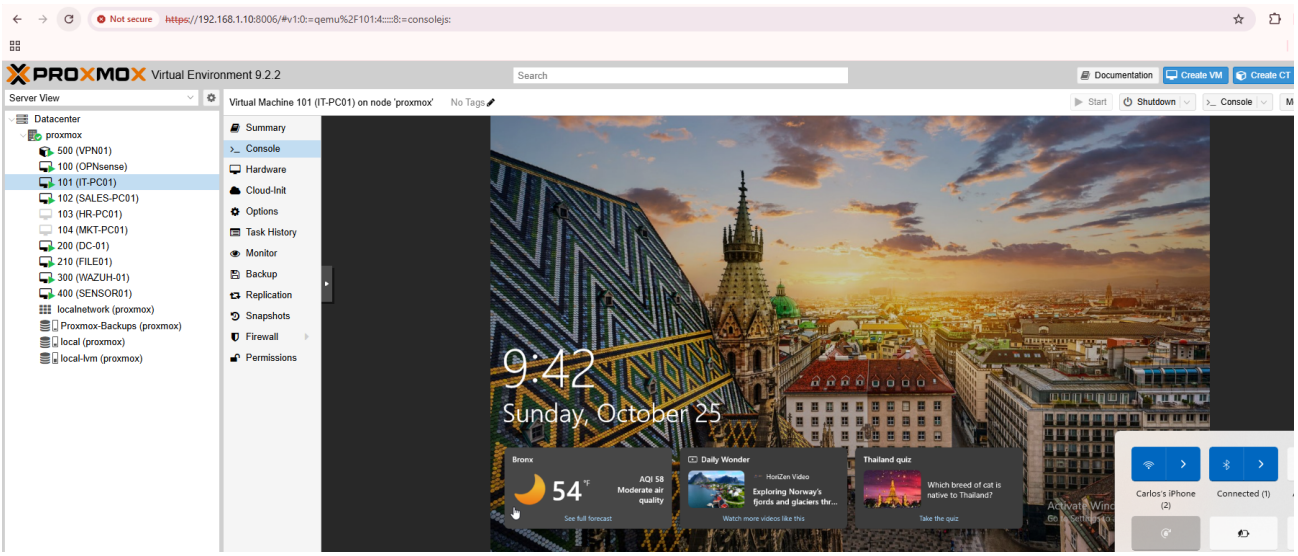


Figure 9. IT-PC01's console works again over the phone hotspot after VPN01 restarts, without re-registering the gateway.

08 / RESULTS AND LESSONS

A working path with clear limits

Validation	Result
Remote dashboard and VM console	Passed over phone hotspot
Laptop disconnect and reconnect	Access failed, then returned
Gateway container reboot	Service and forwarding recovered; remote access returned
Full Proxmox-host reboot	Start at boot configured; recovery not tested

Outcome: REMOTE01 provides a usable remote-management path with a host-scoped route and a narrow service grant. It supports my foundation, networking, and identity-security goals through Linux routing, Proxmox administration, and authenticated device access.

The server, VPN01, and home internet must remain available. Next improvements are a gateway backup, removal of unnecessary nesting after verification, and controlled tests of denied services, an unauthorized device, and full-host recovery.